

Terry's Trousers: E-Commerce Analytics, Strategy & Prompt History

1. Executive Project Overview

Terry's Trousers is a direct-to-consumer (D2C) e-commerce apparel brand specializing in bottomwear. This document consolidates the complete analytical workflow, prompt history, data findings, and tool development completed during our strategic planning session.

Primary Business Objectives

- 1. **Analyze Garment Economics:** Evaluate product line performance across 965 total orders (117,025.00 gross revenue) to uncover margin-draining inefficiencies.
- 2. **Reduce High Return Rates:** Mitigate profit erosion caused by an overall 30.26% return rate (292 returned units costing \$1,172.00 in direct freight/handling).
- 3. **Formulate High-Impact Growth Levers:** Identify the top 3 tactical interventions to maximize net retained profit.
- 4. **Build an Interactive Strategy Tool:** Create a web-based financial model allowing stakeholders to dynamically simulate return reduction and volume expansion scenarios.

2. Chronological Prompt History & Discussion Log

Below is the complete record of user prompts, analytical rationale, and system responses exchanged during this session.

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| PROMPT #1: Initial Business Analysis Request |
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| User Prompt: |
| "I'm an analyst at Fabrix & Novix, a.k.a. an e-commerce business selling trousers. My main goal |
| is to [analyze, e.g., increase my profit this year]. Acting as a business advisor, analyze me |
| [type of data you upload to Gemini] and give me a list of the 3 most important changes I |
| can make that would have the biggest impact on my goal." |
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| File Attached: Trousers Sales Data - Terry's Trousers.csv |
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System Action & Analysis:

- 1. Parsed and cleaned sales dataset (5 product categories, orders, AOV, gross sales, return rates, return fees, and net revenue).
- 2. Identified primary profit drain: **Baggy Cargo** represents 53.8% of revenue but suffers from a 36% return rate (162 units returned, costing \$2,430.00 in processing fees).
- 3. Identified hidden growth champions: **Classic Chinos** and **Condury Stacks** have a low 20% return rate but account for only 21.2% of orders combined.
- 4. Formulated three prioritized recommendations focused on high-revenue retention and low-risk scaling.

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| PROMPT #2: Tool Generation Request |
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| User Prompt: |
| "Based on those insights and my data, create an interactive [type of tool, e.g., profit |
| calculator] for my [type of audience]. Let me define [type of component, e.g., tooltips, |
| sliders, metrics] for each change to show their effect." |
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System Action & Analysis:

- 1. Built a dynamic single-file HTML/JS web application (index.html).
- 2. Embedded live calculation logic linking slider inputs (return rate targets, order volume increases, AOV bundle lifts) to real-time net revenue, handling cost savings, and profit uplift metrics.
- 3. Rendered real-time charts using Chart.js to visually compare baseline performance against simulated scenarios.

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| PROMPT #3: Theme & Aesthetic Refinement |
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| User Prompt: |
| "Can you give this [type of tool, e.g., profit calculator] more of a [specific theme, e.g., |
| outdoors, outdoors]?" |
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System Action & Analysis:

- 1. Redesigned index.html with a bespoke **Aether & Sartorial** Theme.
- 2. Added custom CSS styling including fabric swatch colors, stitched borders (stitched-border), linen background textures, and deep indigo denim headers.
- 3. Updated domain terminology (e.g., fabric types like **Rugged Twill**, **Raw Selvage**, **Raw Denim**, **Cotton Stretch**, **Wide-Wale Corduroy**, plus cut-and-sew COGS recovery controls).

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| PROMPT #4: Documentation Request |
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| User Prompt: |
| "Create a document along with the prompts used in this chat discussion" |
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System Action & Analysis:

- 1. Generated this self-contained Markdown document detailing prompt engineering strategies, financial findings, and tool architecture.

3. Dataset Analysis & Baseline Performance

Baseline Performance Metrics Summary

Overall Return Rate = $\frac{\text{Total Returns}}{\text{Total Orders}} = \frac{292}{965} = 30.26\%$										
Net Retained Revenue = Gross Revenue - Refund Loss - Total Return Costs = \$112,853.00										
Product Line Breakdown Table										
Product Category	Fabric / Style	Orders	AOV	Gross Revenue	Return Rate	Units Returned	Cost / Return	Return Cost	Net Revenue	% Net Rev Share
Baggy Cargo	Rugged Twill	450	\$140.00	\$63,000.00	36%	162	\$15.00	\$2,430.00	\$60,570.00	53.87%
Slim-Fit Denim	Raw Selvage	200	\$110.00	\$22,000.00	28%	56	\$14.00	\$784.00	\$21,216.00	18.86%
Wool Dress Trousers	Fine Flannel	110	\$125.00	\$13,750.00	30%	33	\$13.50	\$445.50	\$13,304.50	11.79%
Classic Chinos	Cotton Stretch	120	\$85.00	\$10,200.00	20%	24	\$12.50	\$300.00	\$9,900.00	8.77%
Condury Stacks	Wide-Wale Cotton	85	\$95.00	\$8,075.00	10%	17	\$12.50	\$212.50	\$7,862.50	6.97%
TOTALS	—	965	\$121.27	\$117,025.00	30.26%	292	—	\$4,172.00	\$112,853.00	100.00%

4. The 3 High-Impact Strategic Growth Levers

Lever 1: Fit & Drapes Optimization for Baggy Cargos

- Root Cause: Baggy Cargos represent 53.8% of sales volume, but loose fit confusion caused a high 36% return rate (162 units returned).
- Action Plan: Implement interactive silhouette/fit guides, 360-degree video drape showcases, and precise waist-to-ankle sizing tables.
- Financial Projection: Lowering return rate from 36% to 20% saves 72 returns, recovering \$10,080.00 in net sales and saving \$1,080.00 in handling costs.

Lever 2: Scale High-Satisfaction Staples (Chinos & Condury)

- Root Cause: Classic Chinos and Condury Stacks feature low return rates (20%), indicating high fit satisfaction, but low order volume (205 total orders).
- Action Plan: Reallocate marketing spend toward these staples and introduce multi-buy 2-pack trouser bundles.
- Financial Projection: Increasing order volume by 50% and AOV by 15% yields an additional \$11,074.25 in top-line net revenue.

Lever 3: Sizing Assistance for Fitted Lines (Denim & Wool)

- Root Cause: Structural lines have high AOV (\$110 - \$125) but suffer 28% - 30% return rates due to rigid waist/high-siting.
- Action Plan: Add customer review filters by height/weight and automated fit recommendation prompts at checkout.
- Financial Projection: A 5% return rate reduction across both lines prevents 15.5 returns, retaining \$1,787.50 in sales and saving \$214.25 in fees.

5. Summary of the Interactive Profit Calculator Tool

The generated single-file web application (index.html) includes the following core components:

- 1. **Preset Scenarios:**
 - Recent Model Line: Reverts all inputs to historical dataset figures.
 - Tailor: Top 3 Changes: Pre-populates recommended operational targets.
 - Raw Growth Target: Simulates aggressive market penetration and maximum return reduction.
- 2. **KPI Header Metrics:** Real-time display of Net Retained Revenue, Return Handling Costs, Overall Return Rate, and Forecasted Profit uplift.
- 3. **Interactive Adjustment Sliders:**
 - COGS & Refurbishment Fee Recovery (0 - \$10)
 - Baggy Cargo Return Rate Target (15% - 36%)
 - Staples Order Volume Increase (0% - 100%) & Bundle AOV Boost (0% - 30%)
 - Slim Denim & Wool Dress Return Rate Targets (15% - 30%)
- 4. **Data Visualization & Export:**
 - Dual Chart.js graphics (Bar Chart for Net Profit Comparison, Doughnut Chart for Category Net Revenue Share).
 - Dynamic Summary Table with live footer totals.
 - One-click clipboard export for executive merchandising reports.

6. Guidelines for Extending Prompt Scenarios

When expanding on this analysis in future Gemini chat sessions, consider using these prompt templates:

- Marketing Campaign Simulation: "Act as an e-commerce growth strategist. Calculate the ROI of spending \$2,500 on customer acquisition for Classic Chinos given our current COGS and return-rate metrics."
- Inventory & Stock Planning: "Based on our Baggy Cargo volume boost and return reduction target, create a recommended fabric ordering schedule (in yards) assuming 2.2 yards of twill per trouser unit."
- Customer Retention & LTV: "Model the 12-month Customer Lifetime Value (LTV) for customers who purchase 2-pack trouser bundles versus single-unit buyers."